

Tutorial (for the week of March 10 -16)

1. Suppose a stationary y_t time series follows an $AR(1)$ process, i.e., $y_t = a_0 + a_1 y_{t-1} + u_t$, where u_t is a white noise, $t = 0, 1, 2, \dots$, then.

(a) What is white noise?

(b) Show that $|a_1| < 1$ so that the model is a convergent model.

(c) Calculate $E(y_t)$ and $Var(y_t)$.

2. Review the key concepts in the time series parts.